Three-Arm Randomized Trial for Pre-Diabetes Report

This report reviews a randomized controlled trial comparing three interventions for pre-diabetic patients intended to reduce progression to full diabetes. The three arms are metformin (met), rosiglitazone (rosi) and lifestyle adjustments of diet and exercise (lifestyle). Each patient's outcome is recorded as success or failure, and the average outcome in each arm is its success rate. The trial included 699 patients with 232 in met, 233 in rosi and 234 in lifestyle (Table 1). To measure uncertainty each arm was resampled with replacement 10,000 times, resampling within the arm so the randomized arm sizes were preserved. The success rate was recomputed for each resample. Pairwise average treatment effects (ATEs) were computed as differences between the bootstrapped success rates. Percentile bootstrap confidence intervals at the 90%, 95% and 99% levels were taken from the empirical distribution of each ATE.

Table 1. Outcomes by treatment arm

Treatment arm	Successes	Failures	Total	Success Rate
Met	112	120	232	0.483
Rosi	143	90	233	0.614
Lifestyle	125	109	234	0.534

Figure 1:

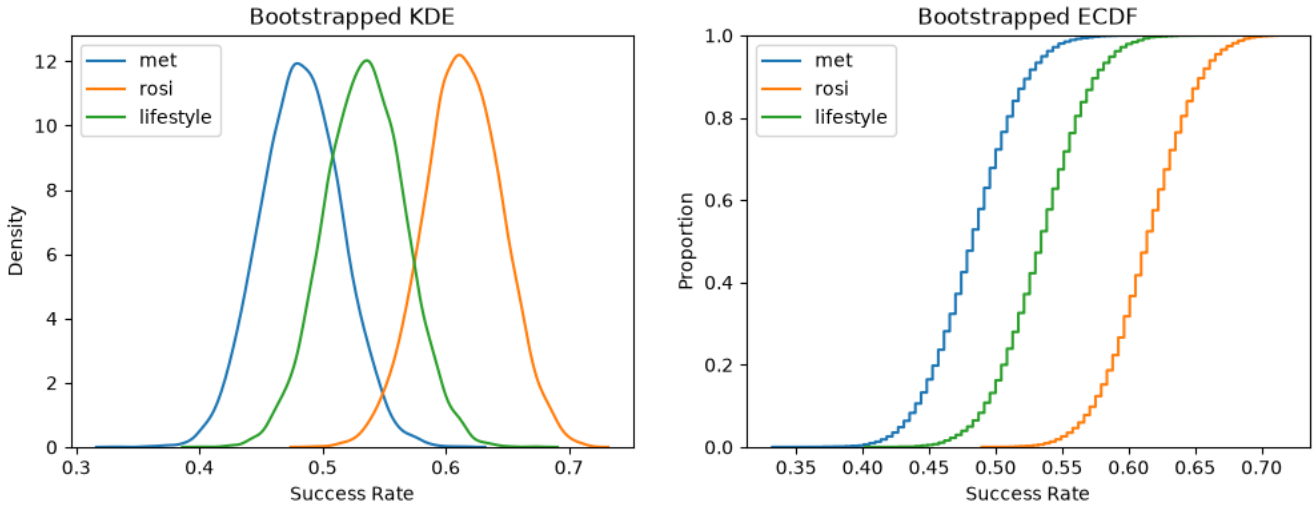
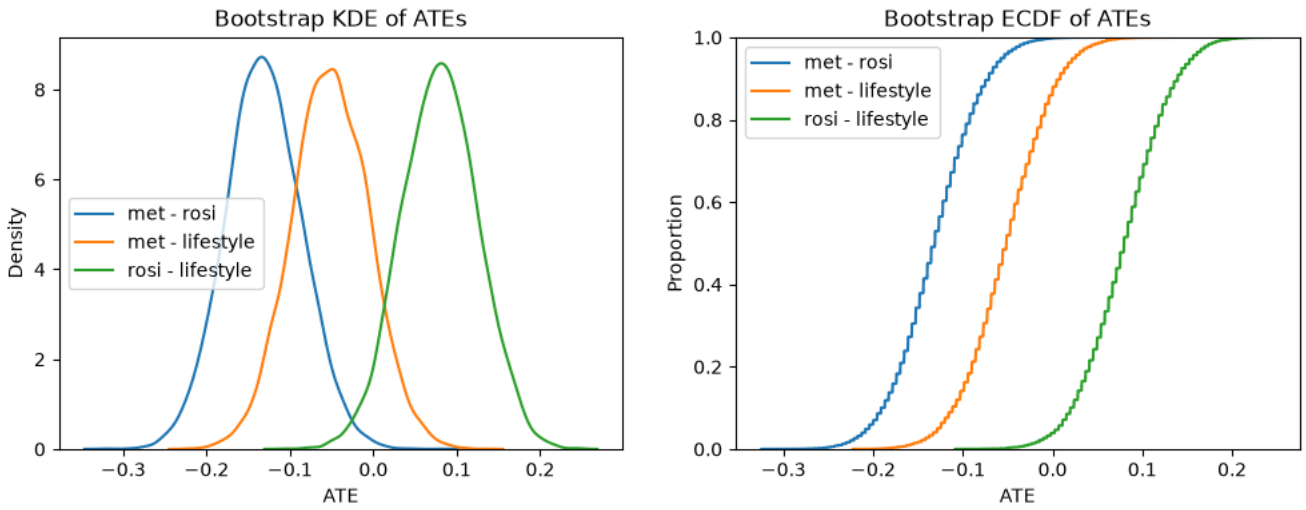


Table 2. Pairwise ATEs with percentile bootstrap confidence intervals

Comparison	ATE	90% CI	95% CI	99% CI	Zero excluded at
Met - rosi	-0.131	-0.204 to -0.054	-0.221 to -0.041	-0.247 to -0.011	90%, 95%, 99%
Met - lifestyle	-0.051	-0.128 to 0.026	-0.141 to 0.039	-0.172 to 0.069	none
Rosi - lifestyle	0.080	0.007 to 0.157	-0.010 to 0.169	-0.036 to 0.195	90%

Figure 2:



Rosi had the highest observed success rate at 0.614 (143 of 233), followed by lifestyle at 0.534 (125 of 234) and met at 0.483 (112 of 232) (Table 1). Met was the only arm with more failures than successes. Figure 1 shows that the bootstrapped success rates in each arm are roughly bell-shaped with similar spreads, centered near the observed rates. The met and lifestyle distributions overlap heavily. The rosi distribution lies to the right of both and overlaps the met distribution only slightly in the tails.

Table 2 and Figure 2 report the pairwise comparisons. The met − rosi ATE is -0.131, and its confidence intervals exclude zero at all three levels (95% CI -0.221 to -0.041). Rosi's success rate is significantly higher than met's. The met − lifestyle ATE is -0.051, and all three of its intervals contain zero (95% CI -0.141 to 0.039), so the trial shows no evidence of a difference between met and lifestyle. The rosi − lifestyle ATE is 0.080. Its 90% interval (0.007 to 0.157) excludes zero, but its 95% interval (-0.010 to 0.169) and 99% interval (-0.036 to 0.195) contain zero. The evidence that rosi outperforms lifestyle is therefore weak and does not reach significance at the 95% level.

Of the three interventions rosi has the strongest evidence of effectiveness in this trial. It has the highest success rate and is significantly better than met at every level tested, although its advantage over lifestyle is significant only at the 90% level. The trial does not support adopting met over lifestyle. Met had the lowest success rate and was not significantly different from lifestyle. Lifestyle fell between the two drugs, and it was not significantly different from met at any level or from rosi at the 95% level.

These conclusions are limited to the single outcome recorded in this dataset. The data contain only each patient's treatment arm and a success or failure outcome. The analysis therefore cannot assess side effects, cost, adherence or differences across patient subgroups, all of which bear on an adoption decision. In addition, three pairwise comparisons were tested without adjustment for multiple comparisons which further weakens the borderline rosi − lifestyle result. Any decision to adopt rosi should weigh this efficacy evidence against safety information that this dataset does not include.